TOPICAL CORTICOSTEROIDS: Comparison Chart

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DRUG/STRENGTH (grouped by formulation & potency)	BRAND NAME	Potency & Cost/30g 1 2 3 4 5 6 7	SIZE / COMMENTS
Potency scale: Ultra-High=1; High=2-3; Mid=4-5; Low=6-7. See Potency Classification* below.		HIGHER … LOWER	
CREAMS → cause less occlusion; suitable for non-acute/wet lesions, intertriginous areas; cosmetically acceptable			
Clobetasol propionate 0.05% *	DERMOVATE, g	$19	15,50,454g / PG
Halobetasol propionate 0.05%	ULTRAVATE ✗ ▼	$50	50g
Betamethasone dipropionate glycol 0.05%	DIPROLENE/TOPILENE GLYCOL PB	$28	15,50g / PG, †
Desoximetasone 0.25%	TOPICORT PB, WA	$34	20,60g / †
Fluocinonide 0.05%	LYDERM, TIAMOL ✗ ▼	$19	100g / PG, †
Betamethasone dipropionate 0.05%	DIPROSONE, g (PG)	$18	15, 50g / †
Beta. dipropionate 0.05% +**clotrimazole** 1%	LOTRIDERM, g	$33	15,50g / OH, PG
Beta. dipropionate 0.05% +**calcipotriol** 50mcg/g	WYNZORA	$68	60g / OH
Amcinonide 0.1%	CYCLOCORT, g (lanolin,paraben,PG,tartrazine,urea free)	$24	30,60g / OH, †
Beclomethasone dipropionate 0.025%	PROPADERM	$26	45,120g / OH
Clobetasate butyrate 0.05% OTC[behind counter]	SPECTRO CREAM ECZEMA *was Eumovate*	$15	30g
Desoximetasone 0.05%	TOPICORT MILD PB, WA	$26	20,60g / †
Mometasone furoate 0.1%	ELOCOM, g (apply **daily**)	$29	15,50g / PG
Triamcinolone acetonide 0.5%	ARISTOCORT C ✗ ▼ *C=concentrate*	$67	15,50g / PG
Triamcinolone acetonide 0.1%	TRIADERM, ARISTOCORT R *R=reg*	$14	30,500g / PG
Betamethasone valerate 0.1% 0.05%	BETADERM, ECTOSONE PB, g BETADERM, ECTOSONE MILD PB, g	$13	15,~450g / † low cost
Betamethasone valerate 0.1% + **fusidic acid** 2%	FUCIBET ✗ ⊗	$88	30g
Hydrocortisone valerate 0.2%	HYDROVAL PB	$17	60g / PG, †
Prednicarbate 0.1%	DERMATOP ✗ ⊗	$88	20, 60g
Desonide 0.05%	DESOCORT, PDP-DESONIDE	$23	60g / PG
Hydrocortisone 1% (OTC) 0.5% (OTC)	HYDERM, POLYSPORIN ECZEMA ANTI-ITCH, g (+ 10% urea JAMP ▼) HYDERM, HYDROSONE, PRAMOX HC, CORTATE, g	$3 $9	15,28,30,45,120 225,454,500g / daily-QID low cost
OINTMENTS → more occlusive, greasy; more effective in dry, scaly, or hyperkeratinized skin areas, palms/soles Avoid in intertriginous areas where they may cause maceration. Cosmetically unacceptable in scalp/hairy areas.			
Betamethasone dipropionate glycol 0.05%	DIPROLENE/TOPICLENE GLYCOL	$28	15,50g / PG
Betamethasone dipropionate 0.05% +SA 3%	DIPROSALIC	$48	50g / psoriasis
Clobetasol propionate 0.05% *	DERMOVATE (PB & lanolin free), g	$19	15,50,454g / PG, †
Halobetasol propionate 0.05%	ULTRAVATE ✗ ▼	$46	50g / PG
Betamethasone dipropionate 0.05%	DIPROSONE, TOPISONE	$18	15,50g
Beta. dipropionate 0.05% +**calcipotriol** 50mcg/g	DOVOBET, g *psoriasis vulgaris; also* ***gel***	$44	60,120g / may ↑Ca
Desoximetasone 0.25%	TOPICORT	$34	60g / PG
Fluocinonide 0.05%	LYDERM	$21	60g / PG
Betamethasone valerate 0.1%	BETADERM	$13	454g / low cost
Mometasone furoate 0.1%	ELOCOM, g (**DAILY** recommended)	$30	15,50g / PG
Triamcinolone acetonide 0.1% 0.1% ***oral topical***	ARISTOCORT R NP ORACORT DENTAL	$16 $66	30g 7.5g / †
Hydrocortisone valerate 0.2%	HYDROVAL PB	$17	60g / PG, †
Betamethasone valerate 0.05%	BETADERM	$12	454g / low cost
Desonide 0.05%	DESOCORT, PDP-DESONIDE	$24	60g
Hydrocortisone 1% (OTC) 0.5% (OTC)	CORTATE, CORTODERM PB	$6 $9	15, 45, 454g / daily-QID low cost

DRUG/STRENGTH (grouped by formulation & potency)	BRAND NAME	Potency & Cost/30g or 60mL 1 2 3 4 5 6 7	SIZE / COMMENTS
Potency scale: Ultra-High=1; High=2-3; Mid=4-5; Low=6-7. See Potency Classification* below.		HIGHER … LOWER	
GELS → non-occlusive, non-greasy, quick drying, & do not leave residue; useful on scalp or hairy areas; irritating			
Beta. dipropionate 0.05% + **calcipotriol** 50ug/g	DOVOBET, g *also* ***ointment***	$48	80g / may ↑Ca**
Desoximetasone 0.05%	TOPICORT	$28	60g / OH
Fluocinonide 0.05%	LYDERM	$21	60g / PG, †
LOTIONS/SOLUTIONS → less occlusive than creams; useful for axilla, foot, groin & hairy areas; & acute weeping lesions			
Betamethasone dipropionate glycol 0.05% lot.	DIPROLENE/TOPILENE GLYCOL	$47	30,60 / PG,OH
Clobetasol propionate * 0.05% scalp lotion 0.05% solution	DERMOVATE SCALP, g CLOBEX SOLUTION, g	$27 $99	60mL 59mL / OH
Halobetasol 0.01% lotion	BRYHALI	$42	60g
Halobetasol 0.01%/**Tazarotene** 0.045% lotion	DUOBRII	$74 ↑irritation	100g
Mometasone furoate 0.1% lotion	ELOCOM, g	$35	30,75 / PG, OH
Betamethasone dipropionate 0.05% lotion	DIPROSONE, TOPISONE	$24	75mL / OH
Betamethasone valerate 0.1% scalp lotion 0.1% lotion 0.05% lotion	BETADERM, ECTOSONE, VALISONE RATIO-ECTOSONE PB RATIO-ECTOSONE MILD PB	$17 $33 $29	75mL / OH 60mL / OH 60mL / OH
Hydrocortisone/**Urea** 1%/10% lotion	DERMAFLEX HC ✗ ▼ *for thickened, rough skin, e.g. elbows*	$20	150mL
Hydrocortisone 1% lotion	PRAMOX-HC[120mL], g[JAMP]	$10	60mL, 120mL
FOAMS/SHAMPOO/SPRAY/OIL → quickly & easily spread; good for hairy areas; alcohol bases may burn if present			
Clobetasol propionate 0.05% topical spray	CLOBEX SPRAY, g ✗ ⊗	$265	59mL / OH
Clobetasol propionate shampoo 0.05%	CLOBEX SHAMPOO, g ✗ ⊗	$139	120mL g, $269[125mL]
Betamethasone dipropionate 0.5mg/g + **calcipotriol** 50mcg/g foam [may ↑Ca++]	ENSTILAR FOAM (shake, spray, & rub in)	$59	60g / Max 100g/wk
Fluocinolone acetonide 0.01% oil	DERMA-SMOOTHE FS OIL ECZEMA DERMA-SMOOTHE FS SCALP OIL ✗ ⊗ DERMOTIC OIL EAR DROPS ✗ ▼	$29 $31 $50 →	118mL / OH 118mL / **Refined peanut oil** **20mL**

Note: **a)** Shaded rows are **common** products for each potency. **b)** **COST** calculated /30g or /60mL using lowest choice

Estimating Amount of Topical Therapy for One Application Using Fingertip Units (FTUs). 1 FTU ≈ 0.5g

Body area *Educate re: correct application amount per site.*	FTU required for one application (Adults)	Amount Needed to Apply BID x 30d
1 hand front/back/fingers or both elbows or both knees or face or genitalia	1	30g
Face & neck	2.5	75g
1 arm or head/scalp	3	90g
Front or back of trunk	7	210g
Buttocks	4	120g
1 leg	6	180g
1 foot	2	60g
Entire body	40	1200g

DOSE ADJUSTMENT

Based on standard growth & development charts[USA] and average BSA (50th percentile).

Infants (<1yr): ≈ 1/5 of adult dose

Children: ≈ 2/5 of adult dose

Adolescents: ≈ 2/3 of adult dose

POTENCY CLASSIFICATION[1,2,3,4] - This 7-point potency scale is based on the US, AHFS and WHO classification system. Potency is based on a human vasoconstrictor assay first used in 1962 which measures visible blanching that correlates with clinical efficacy. Potency scales are not universal across countries. Actual potency may vary considerably depending on: site of application, skin condition, use of occlusion, and individual patient variation. **Propylene glycol (PG)** enhances absorption of the drug, resulting in "ultra-high" potency steroids, which are up to **1000 times** more potent than hydrocortisone. As a general rule, most potent → least potent vehicles include: optimized/augmented vehicles with PG > ointments or gels > creams or lotions.

Potency	Place in therapy	Avoid or Caution	Comments
*Group 1 = Ultra-High Max 50g/wk Group 2, 3 = High Potency Max 100g/wk	✓**resistant conditions**; suitable for **short term** intermittent use in severe eczematous dermatoses & psoriasis. Often required for **palms, soles, scalp & nails** where thickened skin may require prolonged tx, e.g. resistant atopic dermatitis, discoid lupus, hyperkeratotic eczema, lichen planus/sclerosus/simplex chronicus, severe poison ivy, psoriasis, severe hand eczema.	**Avoid:** large areas, thin skin areas, skin folds, face, BSA>30% esp pediatrics.[20] **Caution:** in **young children/infants**	High risk for serious AEs. Apply daily-BID *(once daily often as effective as BID in atopic dermatitis).*[21] Duration: **≤2-4 wks,** then less potent agent.
Group 4, 5 = Mid Potency	✓suitable for intermittent long-term use, chronic use in **thick skin areas**, e.g. anal inflammation (severe), asteatotic eczema, atopic dermatitis, lichen sclerosus (vulva), nummular eczema, scabies (after scabicide), seborrheic/stasis/severe dermatitis, severe intertrigo (short-term), hand eczema.	**Avoid:** thin skin areas **Caution:** face, intertriginous areas	
Group 6,7 = Low Potency	✓safest in **infants**, **children**, & older adults or for covering large or higher risk areas (**face, eyelids, skin flexures, scrotum, perianal**). Suitable for **maintenance** of most chronic conditions after initial control obtained.		Apply BID-QID; less frequent (daily-BID) if ongoing or PRN

For maintenance, try steroid "weekend therapy" (i.e. 2 consecutive days each wk): ↓risk of eczema relapse from 58% to 25% **NNT≈3**.[13] Similar evidence for betamethasone/calcipotriol **ENSTILAR** twice weekly for psoriasis maintenance.[17]

Indications for topical corticosteroids (TCS)

Clinical data supports efficacy in psoriasis, vitiligo, eczema, atopic dermatitis, phimosis, acute radiation dermatitis, & lichen sclerosus. **Limited evidence**: melasma, chronic idiopathic urticaria, and alopecia areata.

CI: untreated skin infections. **Caution** on face & thin skin.

Usual dose: Once to twice daily until clear, then prn.[2x/wk] Use the least potent effective drug for the shortest duration.

AE: skin atrophy, esp if use >6mo;[19] acne; telangiectasias & irreversible **striae** with prolonged use.

Rare: glaucoma/cataracts[eyelids], adrenal suppression / withdrawal reactions[12] esp with high potency +/- occlusive dressing, ↑ area/duration, & **young kids**.

Caution: withdrawal if too sudden/drastic, may cause redness, burning, skin peeling; consider more gradual dose ↓.

Osteoporosis risk[cumulative doses >10,000g in group 1-3 steroids.11]

Concurrent topical non-corticosteroid(s) or systemic agent(s) may ↓ incidence of atrophy from chronic topical corticosteroid use.

SPECIAL POPULATION CONSIDERATIONS

Pediatrics have higher skin surface to body weight ratio than adults and may absorb a higher percentage of topically applied corticosteroids. Diapers/plastic pants on top of treated skin can ↑ absorption up to 10x and ↑ risk of systemic AE. Low potency TCS preferred.

Elderly have thinner/drier skin; ↑ risk of AE (i.e. atrophy, fragility of superficial capillaries[telangiectasia], purpura, bruising, systemic AE). Low potency ointments may be preferred.

PREGNANCY: Use lowest potency (e.g. hydrocortisone or betamethasone valerate) for shortest duration. If potent or very potent TCS is required, suggest <300g during pregnancy due to ↑ risk of low birthweight.

LACTATION: Considered safe as long as high-potency formulations around the nipple is avoided.

Abbreviations: ✗ =non-formulary in SK ▼=covered by NIHB ⊗=not NIHB ✆=prior approval NIHB †=brand specific info in brand section **AE**=adverse events **Beta.**=betamethasone **BID**=twice daily **BSA**=body surface area **FTU**=fingertip unit(s) **g**=gram **lot.**=lotion **OH**=benzyl or isopropyl alcohol (drying!) **NP**=no preservatives **PB**=parabens (preservative; may rarely cause irritation) **PG**=propylene glycol (may rarely cause **irritation** if sensitive) **SA**=salicylic acid **TCS**=topical corticosteroid(s) **WA**=wool alcohol (avoid if wool allergies)

Online Extras for TOPICAL CORTICOSTEROIDS Comparison Chart:

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Eczema in Children – NICE guideline approach http://www.nice.org.uk/guidance/index.jsp?action=byID&o=11636

Tx Escalator ⇨			
			Systemic treatment
			Phototherapy
		Wet dressings	Wet dressings
		Topical calcineurin inhibitors	Topical calcineurin inhibitors
	Low potency corticosteroids	Low potency corticosteroids	Low potency corticosteroids
	emollients	emollients	emollients
	Mild	**Moderate**	**Severe**
	Atopic eczema severity		

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Relative Absorption

Forearm	***1.0***
Sole	***0.14***
Back	***1.7***
Scalp	***3.5***
Forehead	***6.0***
Cheek	***13.0***
Scrotum	***42.0***

Cushing Syndrome (pituitary-adrenal axis suppression)**:**
- 50g of 0.05% clobetasol/wk or
- 500g of 1% hydrocortisone/wk
- in infants: as little as 1g/day x several days may ↓ HPA

Other D/C Products: Beclomethasone dipropionate 0.025% lotion **PROPADERM** (D/C by company); camphor-menthol-phenol emollient lotion **SARNA P** (D/C by company); Clobetasone butyrate 0.05% ointment **EUMOVATE** (D/C by company); Desonide 0.05% lotion **DESOCORT** (D/C by company); Diflucortilone valerate 0.1% + SA 3% **NERISALIC OILY** (D/C by company); Fluocinolone acetonide cream **FLUODERM** (D/C 2005); Halcinonide 0.1% cream, 0.1% ointment **HALOG** (D/C by company); hydrocortisone 0.5% lotion **CORTATE** (D/C 2004); Hydrocortisone/Urea cream 1%/10% **UREMOL-HC** (D/C by company); Triamcinolone acetonide 0.025% **TRIADERM** (D/C 2005) Hydrocortisone 1% lotion, 2.5% cream/lotion, scalp lotion **EMO-CORT;** Diflucortolone valerate 0.1% **NERISONE;** Fluticasone propionate 0.05% **CUTIVATE;** Fluocinolone acetonide 0.01% solution **SYNALAR;** Fluocinolone acetonide shampoo **CAPEX;** Desonide 0.05% Foam **VERDESO**

Search Terms

Amcinonide	32
ARISTOCORT	32
Beclomethasone	32
BETADERM	32
Betamethasone Dipropionate	32
Betamethasone Glycol	32
Betamethasone Valerate	32
BRYHALI	32
Calcipotriol	32
Clobetasol Propionate	32
Clobetasone Butyrate	32
CLOBEX	32
CORTATE	32
Corticosteroid	**32**
CORTODERM	32
CUTIVATE	32
CYCLOCORT	32
DERMAFLEX	32
DERMA-SMOOTHE/FS	32
DERMATOP	32
DERMOTIC	32

DERMOVATE	32
DESOCORT	32
Desonide	32
Desoximetasone	32
Diflucortolone Valerate	32
DIPROLENE GLYCOL	32
DIPROSALIC	32
DIPROSONE	32
DOVOBET	32
DUOBRII	32
ECTOSONE	32
ELOCOM	32
ENSTILAR	32
EUMOVATE	32
Fluocinolone	32
Fluocinonide	32
FLUODERM	32
FUCIBET	32
Halcinonide	32
Halobetasol Propionate	32
HYDERM	32
Hydrocortisone	**32**
HYDROVAL	32
LOTRIDEM	32
LYDERM	32
Mometasone	32
ORACORT DENTAL	32
PRAMOX HC	32
PROPADERM	32
SPECTRO ECZEMA CARE	32
Steroid	**32**
TARO-SONE	32
Topical Corticosteroids	32
TOPICORT	32
TOPILENE GLYCOL	32
TOPISONE	32
TRIADERM	32
Triamcinolone	32
ULTRAVATE	32
WYNZORA	32

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